

Lecture 5 Summary: Actor–Critic Methods

Overview

Actor–critic algorithms combine the strengths of actor-only (policy gradient) and critic-only (value-based) approaches.

- **Actor-only methods** directly optimize policy parameters θ using Monte Carlo estimates of the policy gradient. They can suffer from high variance and lack memory of past estimates.
- **Critic-only methods** (e.g., Q-learning, TD-learning) focus on value-function approximation, indirectly deriving a policy but without direct gradient optimization guarantees.

Actor–critic methods blend both ideas: they estimate value functions (the *critic*) to stabilize and reduce variance in the policy gradient update (the *actor*).

1. Policy Gradient Theorem with Q-Function Approximation

Recall the policy gradient theorem:

$$\nabla_{\theta} \rho(\pi_{\theta}) = \sum_s d^{\pi_{\theta}}(s) \mathbb{E}_{a \sim \pi_{\theta}(\cdot|s)} [(Q^{\pi_{\theta}}(s, a) - b^{\pi_{\theta}}(s)) \nabla_{\theta} \log \pi_{\theta}(s, a)].$$

Instead of Monte Carlo estimates of $Q^{\pi_{\theta}}$, actor–critic methods approximate the Q -function by a parametric model $f_{\omega}(s, a) \approx Q^{\pi_{\theta}}(s, a)$.

Theorem (Sutton et al., 1999). If f_{ω} satisfies the **compatibility condition**

$$\nabla_{\omega} f_{\omega}(s, a) = \frac{1}{\pi_{\theta}(s, a)} \nabla_{\theta} \pi_{\theta}(s, a) = \nabla_{\theta} \log \pi_{\theta}(s, a),$$

and ω minimizes

$$\min_{\omega} \mathbb{E}_{s \sim d^{\pi_{\theta}}, a \sim \pi_{\theta}} \left[\left(Q^{\pi_{\theta}}(s, a) - b(s; \theta) - f_{\omega}(s, a) \right)^2 \right],$$

then

$$\nabla_{\theta} \rho(\pi_{\theta}) = \mathbb{E}_{s, a \sim \pi_{\theta}} [f_{\omega}(s, a) \nabla_{\theta} \log \pi_{\theta}(s, a)].$$

Thus, f_{ω} can safely replace $Q^{\pi_{\theta}}$ in the policy gradient.

2. Examples of Compatible Function Approximators

Softmax policy:

$$\pi_{\theta}(s, a) = \frac{e^{\theta^{\top} \phi_{s, a}}}{\sum_{a'} e^{\theta^{\top} \phi_{s, a'}}}, \quad f_{\omega}(s, a) = \omega^{\top} (\phi_{s, a} - \sum_b \pi_{\theta}(s, b) \phi_{s, b}).$$

The critic thus learns a linear function in the same features as the policy—effectively an approximation of the *advantage function*.

Gaussian policy (continuous actions):

$$\pi_{\theta}(s, a) = \mathcal{N}(\phi(s)^{\top} \theta, \sigma^2), \quad f_{\omega}(s, a) = \frac{(a - \theta^{\top} \phi(s))}{\sigma^2} \phi(s)^{\top} \omega.$$

3. Actor–Critic Algorithm

For each iteration k :

(a) **Policy evaluation:**

$$\mathbb{E}_{s, a \sim \pi_k} [(Q^{\pi_k}(s, a) - b_k(s) - f_{\omega}(s, a)) \nabla_{\omega} f_{\omega}(s, a)] = 0.$$

(b) **Policy improvement:**

$$\theta_{k+1} \leftarrow \theta_k + \alpha_k \mathbb{E}_{s, a \sim \pi_k} [f_{\omega}(s, a) \nabla_{\theta} \log \pi_k(s, a)].$$

This framework updates the critic (value approximation) and actor (policy parameters) simultaneously.

4. Convergence Guarantees (Sutton et al., 1999)

Under standard stochastic approximation conditions:

$$\sum_k \alpha_k = \infty, \quad \sum_k \alpha_k^2 < \infty,$$

and bounded second derivatives of $\pi_\theta(s, a)$, we have:

$$\lim_{k \rightarrow \infty} \nabla_\theta \rho(\pi_{\theta_k}) = 0.$$

In practice, convergence depends on accurate critic training, stable sampling, and small gradient noise.

5. Discussion and Modern Extensions

While theoretically sound, this basic actor–critic scheme is sensitive to approximation errors. Later works addressed instability and efficiency:

- **Kakade & Langford (2002)**: Highlighted instability and proposed trust-region improvements.
- **TRPO (Schulman et al., 2015a)**: Enforces a KL-divergence constraint on policy updates for stability.
- **A3C (Mnih et al., 2016)**: Parallel asynchronous advantage actor–critic.
- **GAE (Schulman et al., 2015b)**: Improves bias–variance tradeoff via temporal discounting of advantages.

6. Key Insight

Actor–critic = **policy gradient** + **learned baseline**. The critic provides low-variance, bootstrapped estimates of Q^π or A^π , while the actor performs gradient ascent using those estimates.